HOMOLOGICAL FUNCTORS WITH FINITE SUMS ARE REPRESENTABLE

THE STACKS PROJECT AUTHORS

Editorial note. Applies the BVdB representability theorem (uncardable, external citation); the duality unwind is in-span.

Allowed standard prerequisites (editorial).

- BVdB representability. theorem-bondal-van-den-bergh (same chapter statement carries an external citation, uncardable here). Represents the twisted functor.
- Perfect duality. cohomology-lemma-dual-perfect-complex (cross-chapter, named). The anti-self-equivalence E mapsto E^vee.

Source reference. Original reference 08DQ: focused-seeds/geometry/S9-eqv-homological-representable/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original theorem 0FYH: sources/stacks-equiv/pinned-original-source.tex, source lines 1032–1032. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let X be a proper scheme over a field k which is regular. Let $G : D_{perf}(\mathcal{O}_X) \to Vect_k$ be a k-linear homological functor such that*

$$\sum\nolimits_{n\in\mathbf{Z}} \dim_k G(E[n]) < \infty$$

for all $E \in D_{perf}(\mathcal{O}_X)$. Then G is isomorphic to a functor of the form $E \mapsto \mathrm{Hom}_X(K, E)$ for some $K \in D_{perf}(\mathcal{O}_X)$.

Proof. Consider the contravariant functor $E \mapsto E^\vee$ on $D_{perf}(\mathcal{O}_X)$, see Cohomology, Lemma 08DQ. This functor is an exact anti-self-equivalence of $D_{perf}(\mathcal{O}_X)$. Hence we may apply Theorem 0FYH to the functor $F(E) = G(E^\vee)$ to find $K \in D_{perf}(\mathcal{O}_X)$ such that $G(E^\vee) = \mathrm{Hom}_X(E, K)$. It follows that $G(E) = \mathrm{Hom}_X(E^\vee, K) = \mathrm{Hom}_X(K^\vee, E)$ and we conclude that taking $K^\vee$ works. □

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